Code: 20A04502T **R20**

B.Tech III Year I Semester (R20) Regular & Supplementary Examinations November/December 2024

DIGITAL SIGNAL PROCESSING

(Electronics and Communication Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

*****

1 Answer the following: (10 X 02 = 20 Marks)

(a)	Compute the output y[n]y[n]y[n] when x[n] = {1, 2, 3} is input into an LTI system with impulse response h[n] = {1,−1} using convolution.	2M
(b)	Analyze whether the system with transfer function $H(z)=\dfrac{z-0.5}{z-1}$ is stable.	2M
(c)	Compute the 4-point DFT of the sequence x[n] = {1, 2, 0, 3}.	2M
(d)	Explain the properties of linear and circular convolution using the sequences x[n] = {1, 2, 3}.	2M
(e)	Compute the transfer function for a digital low-pass IIR filter derived from a Butterworth filter with cutoff frequency ω_c=0.5π.	2M
(f)	Design an IIR filter using the bilinear transformation for a system with a cutoff frequency of 300 Hz and a sampling rate of 1 kHz.	2M
(g)	Compute the FIR filter coefficients for a low-pass filter designed using a Hamming window for a cutoff frequency of 200 Hz and a sampling rate of 1 kHz.	2M
(h)	List the steps involved in designing an FIR filter using the Fourier series method.	2M
(i)	Compute the round-off error for a FIR filter with coefficients {0.5, 0.3, 0.2} quantized to 2 decimal places.	2M
(j)	Mention how quantization of filter coefficients affects the performance of a digital filter.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	Evaluate the time-domain and frequency-domain responses of a discrete-time system with transfer function $H(z)=\dfrac{1-0.5z^{-1}}{1+0.25z^{-1}}$.	10M
	OR	
3	Design a discrete-time system using the difference equation method to have a specific pole at z=0.6 and zero at z=0.3.	10M
4	Evaluate the time complexity of the FFT algorithm for different sequence lengths (N = 4, 8, 16) and compare it with the DFT.	10M
	OR	
5	Demonstrate the process of circular convolution using DFT and verify the result for x[n]={1, 2, 3, 4} and h[n] = {4, 3, 2, 1}.	10M
6	Analyze the pole-zero plot of a second-order IIR filter and discuss its stability for $H(z)=\dfrac{0.5z}{z^2-1.5z+0.7}$.	10M
	OR	

Contd. in Page 2

7	Apply the bilinear transformation to design a high-pass IIR filter with a cutoff frequency of 400 Hz and a sampling rate of 2 kHz. Calculate the filter coefficients.	10M
8	Compare the performance of FIR filters designed using the Hamming and Blackman windows for the same cutoff frequency. Discuss the differences in terms of the main lobe width and side lobe attenuation.	10M
	OR	
9	Analyses the frequency response of a FIR filter designed using the windowing method (Blackman window) and compare it with a FIR filter designed using the rectangular window.	10M
10	Analyses the impact of multi-rate processing (decimation and interpolation) on the aliasing and reconstruction errors in a system with a decimation factor of 4 and an interpolation factor of 2.	10M
	OR	
11	Design a poly phase structure for a decimator with a factor of 3. Derive the poly phase components and compute the decimated output for a sequence x[n] = {1, 3, 5, 7, 9, 11}.	10M

*****